
365
SQL
VS
NoSQL
Essential differences,
schemas, scalability,
use cases & more

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SQL Structured Query Language

Databases are:

- Structured
- Relational & table-based
- Data structure is known in advance
- Relationships between the tables are defined by schema
- Vertically scalable

NoSQL Not Only SQL

Databases are:

- Designed to handle a wide variety of data models like document, key-value, wide-column & graph formats
- Able to manage unstructured/semi-structured data
- Non-relational
- Flexible
- Horizontally scalable

SCHEMA

- Require a **predefined schema** based on the data structure
- The **data structure must be known in advance** and adhered to



- **Dynamic schemas** for unstructured data
- Data can be stored in many ways
- Documents can be created **without predefining** their structure; each can have **its own unique structure**



SCALABILITY

Vertically scalable

- Require more powerful hardware
- The **load on a single server** can be increased by increasing CPU, RAM, or SSD



Horizontally scalable

- Handle more traffic by adding servers in the database and distributing the data across **multiple servers**



CONSISTENCY

- Typically **ACID compliant** (Atomicity, Consistency, Isolation, Durability), which ensures **reliability in every transaction**
- Follow the **CAP theorem** (Consistency, Availability, Partition tolerance), and **focus on performance, scalability, and flexibility** over strict consistency

USE CASES

- Best suited for **complex queries** and transactions where **data structure does not change** frequently
- **Examples**:
  - Financial systems
  - Inventory management
  - Applications requiring complex transactions and precise data integrity



- **Databases**:

MySQL, PostgreSQL, Oracle Database, Microsoft SQL

- Ideal for **large sets of distributed data** and for applications that **require frequent changes** to the data structure, a **flexible schema, and scalability**
- **Examples**:
  - Big data analytics
  - Content management systems
  - Real-time web applications



- **Databases:**

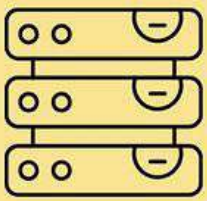
MongoDB, Cassandra, Bigtable, Redis, Couchbase

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DEFINITIONS

SQL

"Structured Query Language"



Databases are:

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- Table-based
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NoSQL

"Not Only SQL"



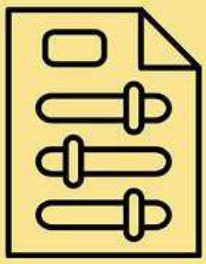
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SCHEMA

SQL



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NoSQL

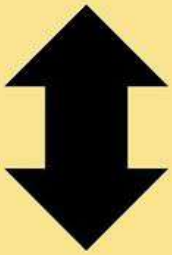


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SCALABILITY

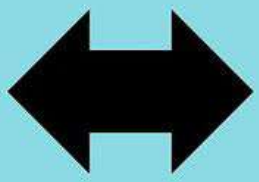
SQL



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NoSQL



Horizontally scalable

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CONSISTENCY

SQL

ACID

- Typically **ACID compliant**
- (Atomicity, Consistency, Isolation, Durability)
- Ensures **reliability in every transaction**

NoSQL

CAP

- Follow the **CAP theorem**
- (Consistency, Availability, Partition tolerance)
- **Focus on performance, scalability, and flexibility** over strict consistency

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USE CASES

SQL



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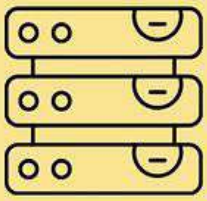
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- **Examples**:
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DATABASES

SQL



- MySQL
- PostgreSQL
- Oracle Database
- Microsoft SQL Server

NoSQL



- MongoDB
- Apache Cassandra
- Google Bigtable
- Redis
- Couchbase